

Dynamic CI Groups and Foundational Group Data

Lab 3.01

⌚ 30m

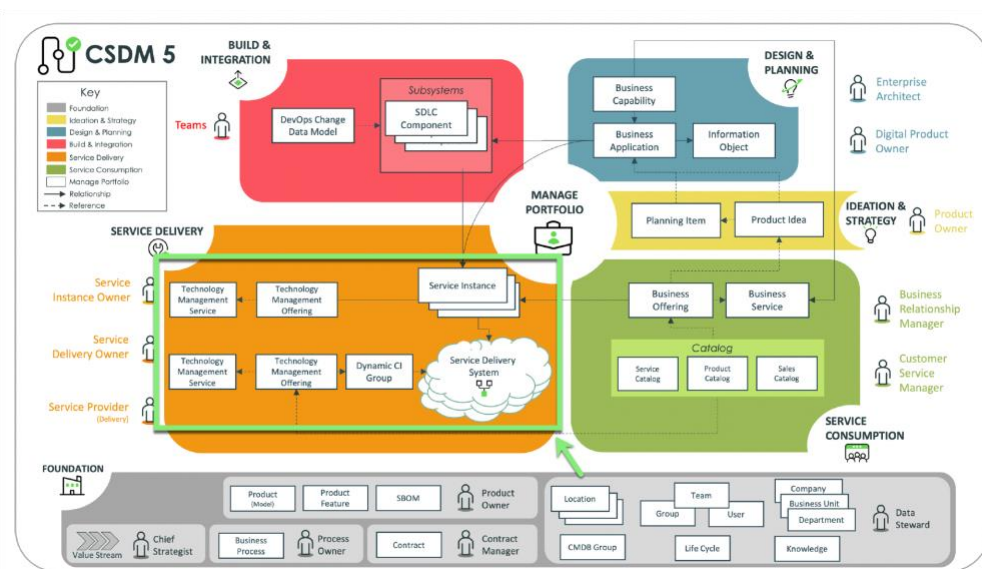
Lab objectives

Dynamic CI groups and the Common Service Data Model (CSDM) are essential for effective IT asset and service management in ServiceNow. Cloud Dimensions has implemented Discovery, Service Mapping, and IntegrationHub ETL to populate the CMDB with rich CI data.

However, CMDB administrators must supplement discovered data with non-discoverable details, such as group assignments. Accurate group data is critical for incident management, change execution, and task assignment.

In this lab, you achieve the following objectives:

- Manage group assignments using CI Class Manager
- Create a CMDB group
- Create a dynamic CI group
- Associate the dynamic CI group to a technology management offering
- Sync group data from the service offering to the underlying CIs



CSDM with focus on areas covered in this lab

A. View Foundational Data

1. Navigate to **Configuration > Servers > Windows**.
2. Search for all servers that start with **SYD** or **CD**.
3. Personalize the list to display the **Support group**, **Change Group**, and **Managed By Group** columns after the **OS Version** column.

The screenshot shows the ServiceNow configuration page for Windows Servers. The search filter is set to 'Name starts with SYD or Name starts with CD'. The table displays the following data:

Name	Operating System	OS Version	Support group	Change Group	Managed By Group
CD-Windows Server	Windows 2012 R2 Standard	6.3.9600	(empty)	(empty)	(empty)
SYD01-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD02-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD03-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD04-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD05-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD06-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD07-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD08-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD09-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD10-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD11-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD12-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD13-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD14-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD15-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD16-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD17-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD18-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)
SYD19-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	Network	(empty)	(empty)

Note: Cloud Dimensions has implemented support groups for some configuration items (CIs) but maintaining them is difficult due to the constant addition of new CIs, which requires manual updates. To address this challenge, Cloud Dimensions plans to use dynamic CI groups and service offerings to simplify ongoing maintenance.

The Support, Change, and Managed by groups are non-discoverable fields that must be manually populated. These fields play a crucial role in processes like incident and change management, and other task assignments. Specifically, the Support group determines the incident assignment group, and the Change group maps to the change assignment group. By implementing dynamic CI groups, Cloud Dimensions can reduce manual effort and improve data accuracy across these processes.

B. Manage Foundational Data from CI Class Manager

Before implementing dynamic CI groups, Cloud Dimensions plans to explore the use of CI Class Manager to populate the Managed By Group field for a specific class of CIs.

1. Navigate to **User Administration > Groups**.
2. Select **New**.
3. In the **Name** field enter, **CD-Managed By Group**.
4. Save the record.
5. Select the **Group Members** related list.
6. Select **Edit**.
7. Search for, select, and move the **System Administrator** to the **Group Members List** pane.
8. Select **Save**.

Note: Associating a group with the Managed By Group field on a CI or service offering helps streamline assigning Data Manager tasks to responsible parties.

9. Reload the form and verify the **System Administrator** appears under the **Group Members** related list.

The screenshot shows the 'Group' form for 'CD-Managed By Group'. The form includes fields for Name, Manager, Hourly rate, and Description. Below these fields are 'Update' and 'Delete' buttons. The 'Group Members' related list is expanded, showing a search bar and a list of users. The 'System Administrator' user is selected and added to the list.

10. Navigate to **Configuration > CI Class Manager**.
11. Select **Hierarchy**.
12. Search for and select the **Windows Server** class.

13. From the **Basic Info** tab, for the **Managed By Group** field, enter **CD-Managed By Group**.

* Table Name	cmdb_ci_win_server
Managed By Group	CD-Managed By Group

Note: The Managed By Group field for all Windows Servers can be configured using CI Class Manager. For more granular control and easier maintenance, you can leverage service offerings and dynamic CI groups, which will be covered later in this lab.

14. Save the record.
15. Navigate to **System Definition > Scheduled Jobs**.
16. Search for and open the **CSDM Data Sync** scheduled job.
17. Select **Execute Now**.

Note: The scheduled job syncs the Managed By Group in CI Class Manager to all CIs within the configured class.

18. Navigate to **Configuration > Servers > Windows**.

Managed By Group
Search
CD-Managed By Group
CD-Managed By Group
CD-Managed By Group
CD-Managed By Group
CD-Managed By Group
CD-Managed By Group

Note: All Windows Servers now reference CD-Managed By Group in the Managed By Group field. This configuration, set through CI Class Manager, applies to the entire CI class. For a more granular and flexible approach, you can configure the Managed By Group using a Technology Management Service Offering in combination with Dynamic CI Groups.

C. Manage Foundational Data from Technology Management Service Offerings

Cloud Dimensions has configured support groups for some of its configuration items (CIs) in the Configuration Management Database (CMDB). However, maintaining these groups is challenging due to the continuous addition of new CIs, which requires manual updates. To address this issue, Cloud Dimensions is exploring the use of dynamic CI groups along with CMDB groups and technology management service offerings.

CMDB groups are collections of CIs organized to facilitate better management, tracking, and maintenance of configuration items and services. Dynamic CI groups automate the inclusion of CIs based on predefined criteria, reducing the manual effort required to maintain these groups and ensuring they are always up to date.

Configure CMDB Group

1. Navigate to **Configuration > CMDB Groups**.
2. Select **New**.
3. For the **Group Name**, enter **CD-Sydney Win Servers CMDB Group**.
4. Save the record.
5. Under the **CMDB Group Contains Encoded Queries** tab, select **New**.
6. Configure the query as follows in the order listed:
 - Class: **Server [cmdb_ci_server]**
 - Condition:
 - **Name | starts with | syd OR Name | starts with | cd**

The screenshot shows the ServiceNow configuration interface for a CMDB Group. The title bar indicates 'CMDB Group Contains Encoded Query' and 'cmdb_ci_server'. The main form includes the following fields and options:

- * Class:** A dropdown menu set to 'Server [cmdb_ci_server]'.
- Domain:** A text field set to 'global'.
- * Group:** A text field set to 'CD-Sydney Win Servers CMDB Group'.
- CI Overview Condition:** Includes buttons for 'Add Filter Condition' and 'Add OR Clause'. Below these are dropdowns for '-- choose field --', '-- oper --', and '-- value --'.
- Condition:** Includes buttons for 'Add Filter Condition' and 'Add OR Clause'. It shows two conditions:
 - Condition 1: 'Name' (dropdown) 'starts with' (dropdown) 'syd' (text field). It is followed by 'AND' and 'OR' buttons, and a red 'X' button.
 - Condition 2: 'or' (radio button) 'Name' (dropdown) 'starts with' (dropdown) 'cd' (text field). It is followed by a red 'X' button.
- Buttons:** 'Update' and 'Delete' buttons are located at the bottom left.

7. Select **Submit**.

8. Select **Show All CI**.

Note: A list of all servers that match the query is displayed.

Create Dynamic CI Group

Dynamic CI groups in ServiceNow automate the inclusion of CIs based on predefined criteria, significantly reducing the manual effort required to maintain these groups. They reference CMDB groups to determine which underlying CIs are managed. Additionally, dynamic CI groups are classified as a type of service in ServiceNow and are stored in the Dynamic CI Group [cmdb_ci_query_based_service] table.

1. Navigate to **Configuration > Dynamic CI Groups**.
2. Select **New**.
3. Configure the form with the following information:
 - Name: **CD-Sydney Win Servers Dynamic CI Group**
 - CMDB Group: **CD-Sydney Win Servers CMDB Group**
4. Select **Submit**.

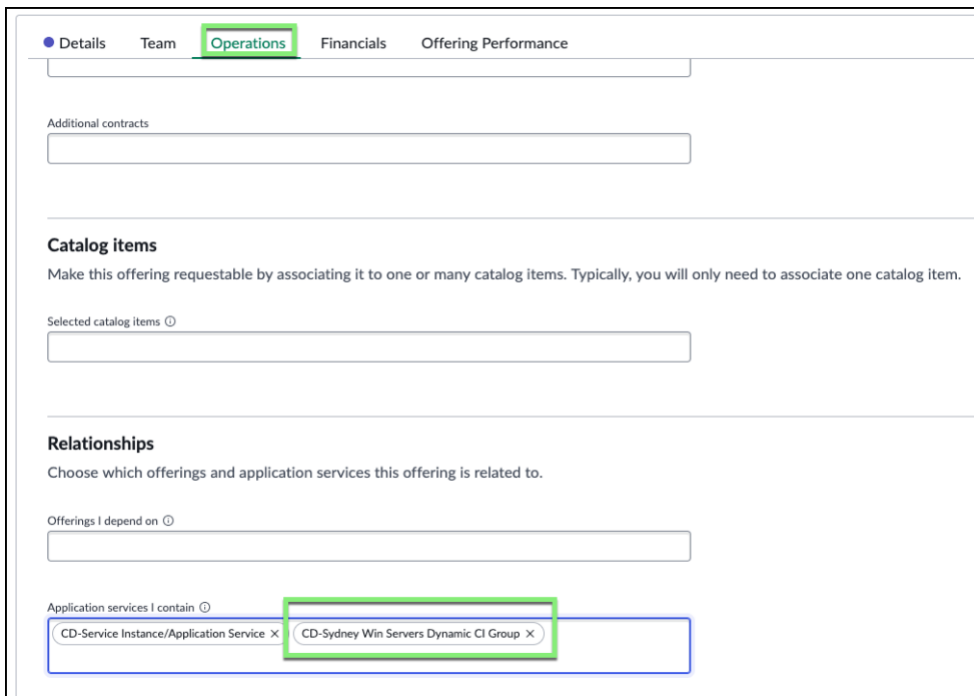
Update Technology Management Service Offering

Groups defined within a service offering are automatically synchronized with the underlying CIs included in the associated dynamic CI group.

Support, change, and managed by groups are non-discoverable fields that must be manually populated on CI records to support processes such as incident and change management. Specifically, the support group maps to the incident assignment group, and the change group maps to the change assignment group fields.

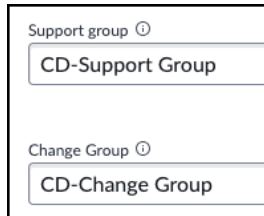
By using technology management service offerings in combination with dynamic CI groups, Cloud Dimensions aims to automate and streamline the management of these groups, ensuring that relevant data is consistently and accurately maintained.

1. Navigate to **Service Portfolio Management > Service Builder**.
2. Under **My services**, select and open **CD-Technology Management Service**.
3. Select the **Manage Offerings** tab.
4. Select and open **CD-Technology Management Service Offering**.
5. Select the **Operations** tab.
6. Scroll to the bottom and in the **Application services | contain** field enter **CD-Sydney Win Servers Dynamic CI Group**.



The screenshot displays the 'Operations' tab of the 'CD-Technology Management Service Offering' in Service Builder. The interface includes tabs for 'Details', 'Team', 'Operations' (selected), 'Financials', and 'Offering Performance'. Below the tabs, there are sections for 'Additional contracts', 'Catalog items', and 'Relationships'. The 'Relationships' section contains a field 'Offerings I depend on' and a field 'Application services I contain'. The 'Application services I contain' field is highlighted with a green box and shows two selected items: 'CD-Service Instance/Application Service' and 'CD-Sydney Win Servers Dynamic CI Group'.

7. Select the **Team** tab.
8. Verify that this offering has already been configured with a support and change group from a prior lab.



The screenshot shows a configuration form with two sections. The first section is labeled 'Support group' with a help icon (i) and contains a dropdown menu with the text 'CD-Support Group'. The second section is labeled 'Change Group' with a help icon (i) and contains a dropdown menu with the text 'CD-Change Group'.

9. Select **Save & Close**.
10. Select the **Review and Submit** tab.
11. Select **Submit**.
12. Select **Return to my dashboard**.

Sync Group Foundational Data to the Underlying CIs

ServiceNow provides a scheduled job that, when executed, synchronizes all group data associated with the service offering to the underlying CIs referenced in the dynamic CI group.

1. Navigate to **System Definition > Scheduled Jobs**.
2. Search for and open **CSDM Data Sync**.
3. Select **Execute Now**.

Note: The scheduled job synchronizes the support group, change group (assignment group), and managed by group from the service offering to the underlying CIs in the dynamic CI group, ensuring efficient management of these non-discoverable attributes.

Outcomes

Leveraging dynamic CI groups delivers several key benefits. When associated with a service offering, they enable streamlined maintenance of foundational group data across the underlying CIs. Additionally, when a dynamic CI group is selected on a change form, ServiceNow automatically unpacks the group, populates the Affected CIs list, and applies the appropriate change assignment groups. This reduces triage time and minimizes the risk of misrouted change requests.

Verify Group Attributes Updated

1. Navigate to **Configuration > Servers > Windows**.
2. Search for all servers that start with **SYD** from the Sydney location.
3. Verify all servers display the updated group values for **Support group**, **Change Group**, and **Managed By Group**.

Windows Servers						
Name	Operating System	OS Version	Support group	Change Group	Managed By Group	
SYD01-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	
SYD02-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	
SYD03-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	
SYD04-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	
SYD05-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	
SYD06-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	
SYD07-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	
SYD08-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group	CD-Managed By Group	

Note: If the Managed By Group set in the CI Class Manager differs from the one configured in the service offering, it will be overridden by the value in the service offering, as it takes higher priority.

Change Management

CI's grouped together in dynamic CI groups benefit Change Management in a few ways:

- When a dynamic CI group is selected as the configuration item on a change request form, its member CIs are automatically unpacked and displayed in the Affected CIs related list.
- The list of CIs in the Affected CIs section can be adjusted to remove any that are not impacted by the change.
- Based on the finalized list of affected CIs, the system can calculate the derived impacted services.

1. Navigate to **Change > Create New**.
2. Under the **Models** tab, select the **Normal** tile.
3. From the **Configuration item** field, select **CD-Sydney Win Servers Dynamic CI Group**.

Note: You previously created this dynamic CI group earlier in the lab, which includes all servers with names starting with SYD or CD. You may remember that the server starting with CD is a downstream CI of the CD-Service Instance/Application Service.

4. Save the record.
5. Verify the **Assignment group** is automatically populated with the **CD-Change Group** as configured from the technology management service offering.

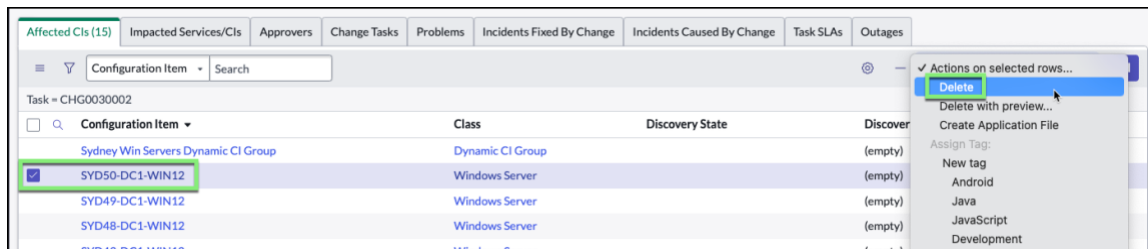
Note: The assignment group on the change form is automatically populated with the change group configured in the technology management service offering. This auto-assignment streamlines the change process by dynamically routing change requests to the appropriate group.

6. Notice the information message stating the unpacking of the group is initiated.

① Unpacking of CD-Sydney Win Servers Dynamic CI Group has been initiated

7. Right-click the header and choose **Reload form**.
8. Under the **Affected CIs** tab, notice all CIs in the dynamic CI group are displayed.
9. To the left of **SYD50-DC1-WIN12**, select the checkbox.

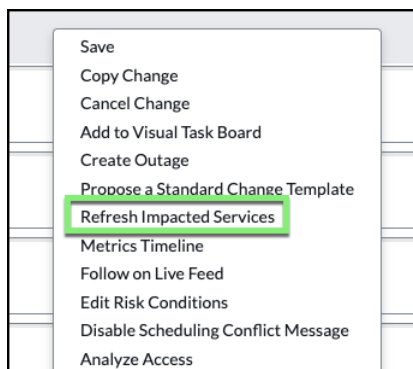
10. From the **Actions on selected rows** choice list, select **Delete**.



11. From the **Confirmation** window, select **Delete** again to remove the CI from the list.

Note: Selecting the dynamic CI group from a change record, provides visibility to all CIs in the group, and allows the change group to manage the change for all the CIs together with the flexibility to remove CIs if they are not needed as part of the change request.

12. From the header, right-click and select **Refresh Impacted Services**.



13. Right-click the header and choose **Reload form**.

14. Select the **Impacted Services/CIs** tab and observe that the change group now has full visibility into the services affected by the change request.

Additional Training: For additional presentations on Change Management and CSDM, view the following video:

- [How Change Management Leverages CSDM](#)

D. Challenge:

Using the skills learned, perform the following tasks:

- Manually create a new **Windows Server** with the following information:
 - Name: **SYD51-DC1-WIN12**
 - Operating System: **Windows 2012 R2 Standard**
 - OS Version: **6.3.9600**

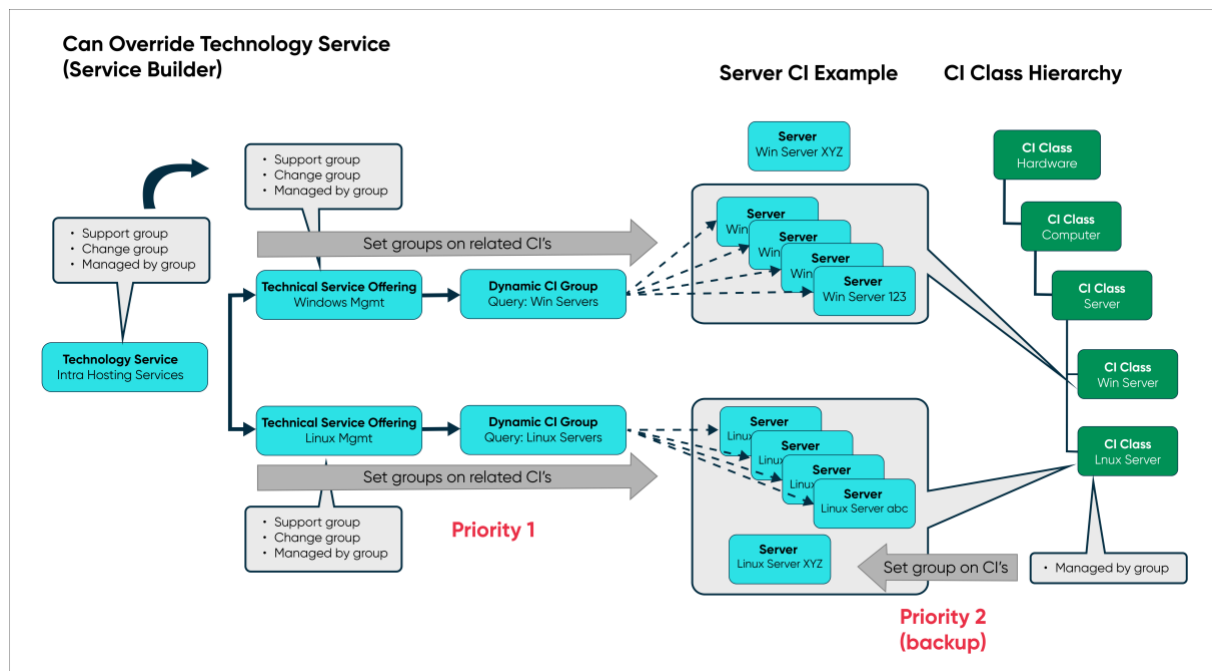
Note: The *Support, Change, and Managed by Group* fields are currently not populated.

- Navigate to **System Definition > Scheduled Jobs** and search for, open and execute the **Update Query Based Services** record. (This job adds up to 100 new CIs at a time that meet the query criteria to the CMDB group and runs every 10 minutes.)
- Navigate to **System Definition > Scheduled Jobs** and search for, open and execute the **CSDM Data Sync** record. (This job syncs the support, change, and managed by group values from the technology management offering to the underlying CIs on daily basis)
- Verify the record you manually created contains all the groups configured from the service offering.

Windows Servers					
Name Search					
Actions on selected rows...					
All	Name	Operating System	OS Version	Support group	Change Group
	Search	Search	Search	Search	Search
	SYD51-DC1-WIN12	Windows 2012 R2 Standard	6.3.9600	CD-Support Group	CD-Change Group
				CD-Managed By Group	

Note: In this example, you manually executed the scheduled job. However, these jobs run automatically according to their defined schedules, ensuring a seamless experience in managing groups and their underlying CIs

In summary, Dynamic CI groups in ServiceNow play a crucial role in enhancing the effectiveness of the Common Service Data Model (CSDM). They help maintain accurate, consistent, and well-organized CI data, which is essential for efficient IT service management and IT operational management. By automating the grouping and classification of CIs, dynamic CI groups contribute to improved visibility, operational efficiency, and alignment with industry best practices.



This diagram illustrates that the Managed by group configured in an offering takes precedence over the Managed by group configured in CI Class Manager for the same CI class.

Lab Complete



Congratulations! You have completed this lab.